

2	(a)	No net force	B1
		No net torque/moment about any point	B1
	(b)	(i) Downwards weight from centre of block	B1
		Force by ball at upper left hand corner of block, follows direction of velocity of ball.	for
		Upwards normal reaction force at A (equal length to weight)	both
		Frictional force acting at A to the left	h
	(ii)	Taking pivot about A, clockwise moments by force from ball = anticlockwise moments by weight.	B1
		$9.0 \times F = 300/1000 \times 9.81 \times 5.0$	C1
		$F = 1.64 \text{ N}$	M1
			A1

3	(a)	(i)	When the gas is heated, the molecules gains kinetic energy (or	B1
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